
COPA: Comparing the incomparable in multi-objective model evaluation

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Abstract

As machine learning (ML) practitioners, we often have hundreds of (trained) ML models at hand from which we need to choose one, based on various objectives such as accuracy, robustness, fairness, scalability, etc. However, how to *compare*, *aggregate* and, ultimately, *trade-off* these objectives is usually a time-consuming task that requires expert knowledge, as they may be measured in different units or scales. In this work, we investigate *how* objectives can be automatically normalized and aggregated to systematically navigate their Pareto front. To do so, we make incomparable objectives comparable using their CDFs, approximated by their relative rankings. As a result, we can aggregate them while matching user-specific preferences, allowing practitioners to meaningfully navigate and search for models in the Pareto front. We demonstrate the potential impact of our approach, named COPA, in both model selection and benchmarking tasks across diverse ML areas such as fair ML, domain generalization, AutoML and foundation models, where classical ways to normalize and aggregate objectives fall short.

1 Introduction

In many phases of machine learning (ML), from model development to deployment, we often need to *compare and select among a population of trained models according to multiple objectives*. For example, even in the simple scenario of a single classification task, model selection involves comparing and selecting among a population of trained classifiers with different hyperparameters to find a specific compromise among objectives such as accuracy, sensitivity, or specificity [24]. A common and more complex scenario these days involves benchmarking a large number of large deep learning models in terms of how they perform with respect to many and diverse objectives that go beyond accuracy, such as robustness [61], fairness [21], and CO₂ footprint [9, 35]. In both examples, we encounter the following challenge: *how do we systematically compare and select among a large number of ML models in terms of multiple objectives?*

Moreover, *different users and applications often have different needs and preferences*. For example, a user may want to download a subset of trained large language models (LLMs) from the *Open LLM Leaderboard* [12] to compare different prompt engineering approaches for a new task. The user requires LLMs that perform relatively well without leaving unnecessarily large CO₂ footprints. To this end, they need to compare the 2148 submitted LLMs in terms of 7 objectives, i.e., their performance across 6 benchmarks and inference CO₂ cost. Among these models, 487 present non-trivial trade-offs, i.e., for every pair, one is better in an objective but worse in another (see Fig. 1). How should they compare the hundreds of models to decide what are acceptable performance-emission trade-offs? Should they manually inspect all 487 LLMs? And what if another user required the most robust model, rather than the most performant? Should they start from scratch?

Similar challenges can be easily found in the literature related to, e.g., multitask learning or domain generalization [43, 48], where the selected model is expected to work ‘well’ on several tasks/domains;

fair classification [62], where it is often unclear what is an acceptable fairness-accuracy trade-off for deployment; or AutoML [15], where tens of frameworks are compared on hundreds of objectives. Crucially, all these works highlight two important limitations in multi-objective ML evaluation:

L1. Objectives with different semantics and domains, such as average performance score and CO_2 cost in Fig. 1, are not directly comparable, and thus cannot be properly aggregated nor traded-off. In physics, this would be akin to comparing meters and grams.

L2. When dealing with many objectives (7 in our LLM example), it is challenging for humans to translate their preferences into a concrete decision, as the number of plausible trade-offs quickly becomes overwhelming (487 in our example).

These challenges reinforce the idea that we need automatic tools to navigate the Pareto front (i.e., the set of optimal trade-offs) in high dimensions, tuning their parameters according to the user preferences. The most common approach would be to perform a weighted combination of either the raw (Naive) or normalized objectives (Norm. and Delta, see §2 for their definitions). However, as we show in Fig. 1, both fail to address L1 and, thus, to evenly explore the Pareto front. In other words, they map most CO_2 importance values, α , to a small region of the front. To overcome these issues, prior works had to devise heuristic approaches tailored to their specific cases [5, 44]. For example, the authors of DecodingTrust [57] had to provide 8 ad hoc rules to normalize their objectives, one per objective. To date, we lack grounded approaches to compare, aggregate and, ultimately, trade-off objectives according to user preferences, that can be used out-of-the-box in multi-objective ML evaluation.

Contributions. We first motivate and establish the incomparability problem in multi-objective ML evaluation, shedding light on why previous approaches fail (§2). Next, we introduce COPA, a novel approach to allow practitioners to meaningfully navigate the Pareto front, and thus compare and select models that reflect their preferences (§3). COPA accomplishes this goal with two components: i) a normalization function that universally makes all objectives comparable via their cumulative distribution functions, which we approximate using relative rankings; and ii) a criterion function with two easily interpretable parameters controlling the aggregation and importance of each objective. We then place COPA in the context of related work (§4), and finally demonstrate its potential impact (§5) in diverse and timely applications such as domain generalization, multitask learning, fair ML, AutoML benchmarking, and LLM selection. As we illustrate in Fig. 1, COPA enables thoroughly exploring the Pareto front as a function of the user preferences, here controlled by α . For instance, a deployer equally interested in the performance and CO_2 emissions of the LLM, could use COPA with $\alpha = 1/2$ to pick the model in the middle of the Pareto front (last row in Fig. 1), ranked top 18% for both objectives.

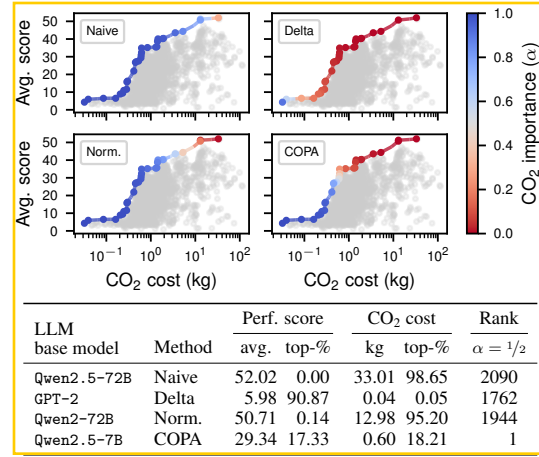


Figure 1: COPA meaningfully navigates the performance-emissions trade-off of the Open LLM Leaderboard [12], evenly mapping the importance of CO_2 cost to the Pareto front. In contrast, existing approaches are biased toward one of the objectives. This is reflected in the retrieved LLMs where, e.g., COPA maps $\alpha = 1/2$ to a top 18% model for both objectives, and all other approaches select either a high-performing but CO_2 -demanding model, or vice versa.

2 Problem statement

We are given a population of already-trained models $\mathcal{H}$, typically obtained by changing hyperparameters, where each model $h \in \mathcal{H}$ is associated to a vector of K metrics assessing its performance with respect to different evaluation objectives. In addition, we assume each objective to be a continuous random variable for which we have sampled observations in $\mathcal{H}$.

Without loss of generality, we assume that each individual objective has to be *minimized*, and we can thus frame the problem as a multi-objective optimization (MOO) problem of the following form:

$$\min_{h \in \mathcal{H}} \mathbf{y}(h) := [y_1(h), y_2(h), \dots, y_K(h)], \quad (1)$$

where $\mathbf{y}(h)$ is the objective vector of model h , and $y_k(h)$ its performance on the k -th objective. When it is clear from the context, we will omit the argument and write $\mathbf{y}$ and y_k directly.

How can we minimize a vector? A fundamental problem of Eq. 1 is that *minimizing the vector $\mathbf{y}$ is not well-defined*, as there is no canonical total order in high dimensions. Therefore, two models could yield objective vectors where one is not always better than the other for all objectives. In the MOO literature, the set of optimal trade-off solutions is known as the *Pareto front* and, more formally, an objective vector $\mathbf{y}^*$ is in the Pareto front (and called *Pareto-optimal*) if there exists no other feasible vector $\mathbf{y}$ such that $y_k \leq y_k^*$ for all $k \in \{1, 2, \dots, K\}$, and $y_k < y_k^*$ for at least one of the objectives. While the Pareto front is theoretically appealing, in practice, the **decision maker** (DM) needs to navigate the Pareto front and, eventually, select one single model.¹ In other words, the DM needs to specify a total order in Eq. 1 which implies: i) taking a total order directly in $\mathbb{R}^K$, e.g., the lexicographic order where $\mathbf{y} < \mathbf{y}^*$ iff $y_k < y_k^*$ and $y_i = y_i^* \forall i < k$; or ii) defining a **criterion function** $C: \mathbb{R}^K \rightarrow \mathbb{R}$ to rewrite Eq. 1 as a scalar-valued problem:

$$\min_{h \in \mathcal{H}} C(\mathbf{y}(h)). \quad (2)$$

One remarkable example of the latter is the *global-criterion method* [63] which maps DM preferences to the problem geometry by interpreting Eq. 2 as selecting the model closest to the *ideal* one, i.e.,

$$\min_{h \in \mathcal{H}} \|\mathbf{y}(h) - \mathbf{y}^{\text{ideal}}\|_*, \quad (3)$$

where $\mathbf{y}^{\text{ideal}}$ is the ideal solution, $\mathbf{y}^{\text{ideal}} := [\min_h y_1, \min_h y_2, \dots, \min_h y_K]$ and $\|\cdot\|_*$ is typically a p -norm. However, naively solving Eq. 3 (and, more generally, Eq. 2) is well-known in the MOO literature to be sensitive to the scaling of the objectives [4] (recall **L1** in §1), and thus prevents us from properly accounting for any DM preferences (**L2**). In this work, we argue that the criterion function C should fulfill the following desiderata:

- D1.** Reflect the DM preferences, translating their model expectations into an optimization problem
- D2.** Provide a simple way to tune its parameters to meaningfully explore the Pareto front.

When are objectives incomparable? Similar to dimensional analysis in physics [2]—which argues that we cannot combine incommensurable quantities, e.g., kilograms and meters—we argue that a second fundamental issue that we face in Eq. 2 is **semantic incomparability**, i.e., whether it is sensible to compare (and thus aggregate) the values of two different objectives.

For example, if objectives differ in their semantics they are hardly comparable in general, e.g.: despite both accuracy and ROC AUC lying in the unit interval, it does not make immediate sense to compare their values. There are, however, other aspects that are more subtle. To illustrate these, Fig. 2 presents a synthetic Pareto front from §5.1 where both objectives quantify prediction error in significantly different domains, namely, within the intervals $[0, 0.2]$ and $[0.5, 3.0]$. We navigate the Pareto front solving a weighted Tchebycheff problem [3] of the form

$$\min_{h \in \mathcal{H}} \max \{ \alpha |y_1|, (1 - \alpha) |y_2| \}, \quad (4)$$

which solves Eq. 3 with C as the ∞ -norm weighted by $\alpha \in [0, 1]$. Intuitively, Eq. 4 looks for robust solutions that account for the importance of solving one objective over the other, seemingly satisfying our desiderata **D1-2**. However, its naive application over the original objectives clearly shows how we can bias model selection in favor of Objective 2, as it can be seen in Fig. 2: for any given preference α smaller than 0.75, Eq. 4 yields a solution which *completely ignores Objective 1 performance*.

How can we make objectives comparable? As we just discussed, *even if we use a well-designed criterion function*, semantic incomparability can hinder our goal to meaningfully explore the Pareto

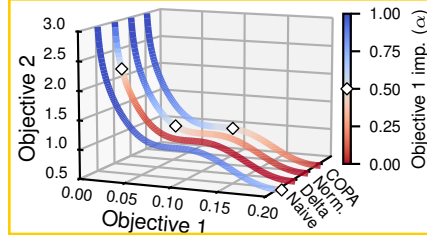


Figure 2: As we explore a synthetic Pareto front with different normalization functions to solve Eq. 3, only COPA meaningfully navigates it as we change α , and its min-max solution agrees with our expectations of a robust solution.

¹Note that, when we plot the Pareto front in 2D, e.g., in Fig. 1, the linear interpolation between models (dots) only serves visualization purposes, i.e., we cannot interpolate between models.

front. Historically, this has been addressed in the MOO literature by applying **component-wise transformations** to the objectives to normalize them [40], turning Eq. 2 into

$$\min_{h \in \mathcal{H}} C(\phi(\mathbf{y})) := C([\phi_1(y_1), \dots, \phi_K(y_K)]) . \quad (5)$$

Two classic examples of these transformations are

$$\Delta_k(y_k) := \frac{y_k - y_k^{\text{ideal}}}{y_k^{\text{ideal}}} \quad \text{and} \quad \text{norm}_k(y_k) := \frac{y_k - y_k^{\text{ideal}}}{y_k^{\text{nadir}} - y_k^{\text{ideal}}} , \quad (6)$$

where $y_k^{\text{nadir}} := [\max_h y_1, \max_h y_2, \dots, \max_h y_K]$ is the worst possible solution. Intuitively, Δ_k represents the difference relative to the ideal solution, and norm_k reweighs the objective to lie in the unit interval. Prior works have extensively used Δ_k , often replacing y_k^{ideal} with a reference vector, as computing it can be challenging [33, 38, 40]. Back to our synthetic case, we now want to solve

$$\min_{h \in \mathcal{H}} \max \{ \alpha |\phi_1(y_1)|, (1 - \alpha) |\phi_2(y_2)| \} . \quad (7)$$

By testing different ϕ_k , we can understand why classic approaches fail to make objectives comparable. More specifically: **i)** using Δ_k now biases the problem toward the first objective, since $\min_h y_1 \approx 0$; and **ii)** using norm_k alleviates these problems, as the denominator is now bigger than the numerator, yet the differences between distributions (that of y_k being heavy-tailed) still bias the optimization towards the first objective. Instead, we seek to explore the Pareto front making a more meaningful use of α , spreading it uniformly along the curve.

The *main goal* of the functions $\phi_k: \mathbb{R} \rightarrow \mathbb{R}$ is thus to make the objectives semantically comparable, so that we can seamlessly aggregate them with the criterion function C . To this end, we argue that the functions ϕ_k should be:

D3. Objective-agnostic, so that we can normalize any objective irrespectively of its specific nature.

D4. Order-preserving (i.e., strictly increasing), so that it preserves Pareto-optimality.

In summary, to meaningfully explore the Pareto front, it is important to design a criterion function C that translates well DM preferences into an optimization problem (D1-2), and a normalization function ϕ that makes objectives semantically comparable (D3-4). These desiderata will blend in COPA, discussed in the next section. In the synthetic experiment above, COPA maps the value $\alpha = 1/2$, which turns Eq. 7 into a robust min-max problem [56], to the flat region of the curve in Fig. 2, matching the intuition of what a robust solution should represent.

3 Methodology

Next, we introduce the proposed normalization and criterion functions fulfilling the desiderata D1-4 described in §2. We refer to the problem resulting of solving Eq. 5 with the proposed functions as *cumulative-based optimization of the Pareto front* or, in short, **COPA**.

3.1 Designing a universal normalization function

We argued in §2 that the function ϕ should fulfill desiderata D3-4, i.e., it should make any objectives semantically comparable while preserving their Pareto-optimality. Taking advantage of our probabilistic perspective (recall that y_k is a continuous random variable), we propose to design ϕ such that the resulting variables are all equally distributed and, w.l.o.g., uniformly distributed in the unit interval. That is, we propose to use $\mathbf{u} := [u_1, u_2, \dots, u_K]$ instead of $\mathbf{y}$, where

$$u_k := F_k(y_k) \sim \mathcal{U}(0, 1) \quad \forall k \in \{1, 2, \dots, K\} , \quad (8)$$

and $\phi_k = F_k$ is the marginal cumulative distribution function (CDF) of the k -th objective. Indeed, this transformation is known in statistics as the probability integral transform [6, Example 5.6.3], and u_k is guaranteed to follow a standard uniform distribution if y_k is continuous.

Remarkably, Eq. 8 makes all criterion functions *marginal-distribution-free* in the sense of Kendall and Sundrum [29], i.e., it strips away all individual properties of the marginal distributions (e.g., the domain) of any given objective (D3). We note that normalizing random variables this way is one of the fundamental building stones of copulae in statistics [14, 51], ensuring that copula functions exclusively learn the relationship across random variables.

How can we interpret the values of $\mathbf{u}$? One important advantage of using $\mathbf{u}$ in place of $\mathbf{y}$ in Eq. 5 is that it provides a common framework to think about all objectives, since all their values are now framed as *elements within a population*. In practice, this means that the DM has a common language to express their expectations on the model. For example, $\mathbf{u} = \mathbf{1}/2$ corresponds for all objectives to the *the median value*, which divides $\mathcal{H}$ into two *halves* comprising the best and worst performing models.

However, there is still one caveat we need yet to address: we have no access to the marginal CDF of each objective, but only to samples of the joint distribution in $\mathcal{H}$.

3.2 Rankings as finite-sample approximations

As mentioned above, while we have no access to the CDFs themselves, we have samples from the joint distribution over the objectives, i.e., over $p(y_1, y_2, \dots, y_K)$. Namely, we can consider each model $h \in \mathcal{H}$ as a sample from the joint distribution and, by looking at each objective individually, as a sample from the marginal distributions.

Let us now focus on the k -th objective y_k , and drop the subindex in the following to ease notation. Say that we have $|\mathcal{H}| = N$ i.i.d. realizations of the objective, i.e., $\{y_1, y_2, \dots, y_N\} \stackrel{\text{i.i.d.}}{\sim} P_k$. Then, we can approximate Eq. 8 for the i -th sample, $u_i = F(y_i)$, by computing its order statistic, i.e., the random variable representing its relative ranking within the population, $R(i) := \sum_{j=1}^N [y_j < y_i]$ where Iverson brackets denote the indicator function, such that $y_{R(1)} \leq y_{R(2)} \leq \dots \leq y_{R(N)}$. Specifically, since the *empirical CDF* is the fraction of samples smaller than the input, it is direct to show that

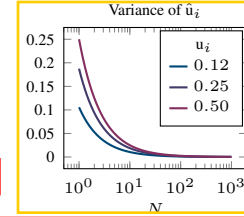
$$\hat{u}_i = \hat{F}(i) := \frac{1}{N} \sum_{j=1}^N [y_j < y_i] = \frac{1}{N} R(i) \quad (9)$$

enjoys the following properties [6]:

Proposition 3.1. $\hat{u}_i$ is an unbiased estimator of the CDF at y_i , u_i , with variance $u_i(1 - u_i)/N$. The variance of $\hat{u}_i$ decreases linearly with N , and has a maximum value of $0.25/N$ at the median.

Proof. First, note that $[y_j < y_i] \sim \text{Bern}(u_i)$. Then, we have $R(i) \sim \text{Bin}(N, u_i)$ with mean Nu_i and variance $Nu_i(1 - u_i)$. Hence, $\hat{u}_i$ has mean $\frac{1}{N} \mathbb{E}[R(i)] = u_i$, and variance $\frac{1}{N^2} \mathbb{V}[R(i)] = u_i(1 - u_i)/N$ which, by taking derivatives w.r.t. u_i , $\partial_{u_i} \mathbb{V}[\hat{u}_i] = 1 - 2u_i = 0 \Rightarrow u_i = 1/2$, which is a maximum since $\partial_{u_i}^2 \mathbb{V}[1/2] < 0$. $\square$

In other words, we can use the relative rankings of each objective to build an unbiased² estimator of the CDF, $\hat{u}_i$, whose variance rapidly decreases as we increase the size of $\mathcal{H}$, i.e., $\mathbb{V}[\hat{u}_i] \rightarrow 0$ as $N \rightarrow \infty$. Indeed, the inset figure shows the variance of $\hat{u}_i$ as a function of the sample size for three different values of u_i . Note that the relative ranking is strictly increasing: if $y_i < y_j$, then $F(y_i) < F(y_j)$ for any $\mathcal{H}$ containing both samples (D4). While this is an approximation of the true CDF, which would retain instead all the information about the joint distribution, it works egregiously well in our experiments (§5). Furthermore, note that this transformation is meant to ease inter-objective computations, we can (and should) use the original values of y_k to perform intra-objective comparisons or decisions.



3.3 Incorporating preferences into the optimization

Now that we can effectively approximate our normalization function, we introduce a criterion function to translate DM preferences into an optimization problem (D1). To do so, we start by looking back at global criterion methods, since plugging in our transformation $\mathbf{u} = \phi(\mathbf{y})$ simplifies the problem in Eq. 3 to $\min_b \|\mathbf{u}\|_*$ as the ideal point becomes the origin, i.e., $\mathbf{u}^{\text{ideal}} = \mathbf{0}$. Then, by using the approximation described in §3.2, the problem becomes a simple finite search of the form

$$\min_{i \in \{1, 2, \dots, N\}} \|\hat{\mathbf{u}}_i\|_* \quad (10)$$

That is, we have reduced our problem to finding the model whose ranking vector has the smallest norm. Using this new *marginal-free global-criterion method*, mapping the DM preferences now boils down to selecting an appropriate norm for the problem in Eq. 10. To this end, we propose to use as criterion function C a norm with parameters $p \geq 1$ and $\omega \in \mathbb{R}_+^K$ defined as

²In fact, it is known to be a consistent estimator [55].

$$\|\mathbf{u}\|_{p,\omega} := \left(\sum_{k=1}^K |\omega_k \mathbf{u}_k|^p \right)^{1/p}, \quad (11)$$

where $\sum_k \omega_k = 1$. This norm can be interpreted as a regular p -norm on a space with coordinates scaled by ω . More remarkably, note that this differs from the usual weighted p -norm, as the weights are *inside* the absolute value. We justify this choice given that the values of $\mathbf{u}_k$ lie in the unit interval, and the power would often make them vanish too quickly, as we demonstrate in Fig. 10.

How can we interpret the parameters? Fortunately, the parameters of the proposed criterion function, p and ω , provide an easy and interpretable way for the DM to navigate the Pareto front (D2). Regarding ω , as we apply them in Eq. 11 *before* taking the power, we can provide a clear interpretation of ω in terms of ratio trade-offs. For example, if we had two objectives with $\omega = [0.75, 0.25]$, then we can see by equating the weighted objectives that minimizing the first objective to a value of $\mathbf{u}_1$ is worth the same as minimizing the second objective to a value of $\mathbf{u}_2 = \omega_1 / \omega_2 \mathbf{u}_1 = 3\mathbf{u}_1$, i.e., $\mathbf{u}_1$ is three times more important than $\mathbf{u}_2$. If we combine this interpretation with that of $\mathbf{u}$ given in §3.1, we could say, e.g., that we value being in the top 25% of the models for the first objective the same as being in the top 75% for the second objective.

We can interpret p using the same intuition as in ML regularization [16]: the models selected in Eq. 10 are those first intersecting an ever-expanding p -ball centered at the origin, whose shape depends on p . Higher values of p lead to denser objective vectors, while smaller values lead instead to sparser ones. Additionally, some values of p have clear interpretations: $p = 1$ is the average rank; $p = 2$ is the Euclidean distance; and $p = \infty$ turns Eq. 10 into a min-max problem, typically used to formulate robust optimization problems [56].

Does Eq. 11 enjoy theoretical guarantees? Given the similarity with commonly-used norms, it is natural to ask whether we can leverage existing results from the MOO literature and adapt them to the proposed norm. This is indeed the case, and we can easily guarantee, e.g., that the solutions found using Eq. 11 with $1 < p < \infty$ are always Pareto-optimal [40, Thm. 3.4.1]. However, it might not reach all optima. Similarly, note that $p = \infty$ reduces Eq. 10 to a weighted Tchebycheff problem which reaches any Pareto-optimal solution [40, Thm. 3.4.5], but also weakly optimal ones.

In practice, using a weighted Tchebycheff problem ($p = \infty$) is a good practice when we have few objectives and a large budget for the weights ω to test. Instead, when interested in finding a particular model (i.e., solving Eq. 5 once), we suggest setting p based on the level of robustness desired (as lower values of p lead to higher tolerance to bad performance on individual objectives), and ω based on the importance of solving each objective given by the DM.

4 Related work

Our work draws connections with other scientific domains, e.g., the notion of semantically incomparability is akin to that of incommensurability in dimensional analysis [2]. Similarly, using relative rankings to make better comparisons has been previously explored in microeconomics [47], MOO [23, 31], and statistics, designing methods that avoid the normality assumption, e.g., the Friedman test [13], Wilcoxon signed-rank test [60], or Kendall's τ coefficient [28]. Finally, as mentioned in §2, copulas exploit the probability integral transform to become marginal-distribution-free [14], and the proposed criterion functions share similarities with weighted L_p -problems in MOO [40].

In ML, the closest work to ours is Park et al. [46], which learns the joint CDF, approximated with a copula, to recover a partial order for multi-objective Bayesian optimization. In contrast, we employ marginal CDFs and provide a principled way to translate DM preferences to an optimization problem. Another line of related works are those that attempt to learn the Pareto front either for model merging [7, 32] or a posteriori MOO methods [64]. Unfortunately, these methods fail to address semantic incomparability as they use the raw objectives. ROC curves [11] provide an interesting connection, since their axes can be understood as the CDFs of the target classes [18]. In practice, many prior works proposed ad hoc approaches to normalize and aggregate objectives using, e.g., normalized RMSEs [44, 57]—we refer to §8.3 of Japkowicz and Shah [24] for other references. Notoriously, some works in multitask learning [33, 43] and domain generalization [48] use rank averages to aggregate objectives, yet the standard is to use the average of Δ_i normalized objectives (see Eq. 6). COPA can benefit these two areas, along any others accounting for several objectives such as fair ML [39], federated learning [27], probabilistic ML [26], and multimodal learning [1].

5 COPA in action

In this section, we motivate the use of COPA by showing a range of practical scenarios which would benefit from adopting the proposed methodology. We defer additional details and results to §A.

5.1 Synthetic evaluation

To qualitatively assess COPA, we consider a synthetic Pareto front of the form $y_2 = 0.25 \cos(39y_1^{0.85}) - \log(y_1) - 0.46$ where $y_1 \sim \mathcal{U}(0.02, 0.2)$. We obtain as a result a non-convex Pareto front with a flat area around $y_1 = 0.1$, and two objectives with significantly different distributions.

Does the parameter p match our intuitions?

We corroborate the insights from §3.3 by showing in Fig. 3 the distribution of solutions found taking different values of p . First, note that since the front is *strictly* increasing except in $[0.083, 0.091]$, we have that $u_1 \approx 1 - u_2$. As a result, we see that $p = 1$ almost exclusively finds solutions on the extrema. When we increase p , the distribution of solutions better spreads along the front and, as the p -balls become more square-like, we gain finer control on the solution found by tuning α . It is important to stress, however, that the finer control of $p = \infty$ comes at cost: as we increase K , finding a proper ω could prove challenging.

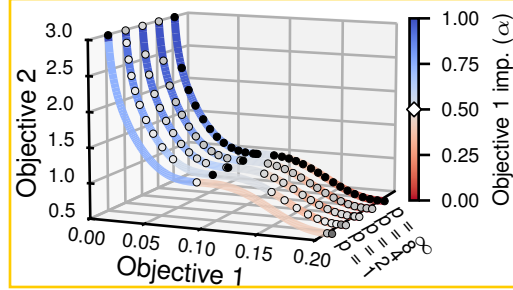


Figure 3: Distribution of solutions (circles) found for different values of p as we sweep over values of α . The darkness of the circles represents the number of times they were selected.

5.2 Case 1: Model selection

First, we explore how the norm proposed in §3.3 can help us explore the Pareto front more meaningfully, i.e., how sensibly it maps the DM preferences to Eq. 5:

1. The performance-emissions trade-off. Despite LLMs recently showing outstanding performance [42], their CO_2 footprint can be concerning and needs to be taken into account [9]. Next, we show how practitioners can leverage COPA to better navigate this crucial trade-off in the LLM space.

We gather the results of 2148 LLMs submitted to the Open LLM Leaderboard [12] and take as objectives their inference CO_2 cost and performance on 6 different datasets: IFEval [65], BBH [54], MATH [20], GPQA [49], MuSR [52], and MMLU-Pro [58]. Then, we use COPA with $p = \infty$ to select an LLM, changing ω as we vary the importance given to their CO_2 footprint, denoted by α , as $\omega := [\alpha, (1-\alpha)/6, \dots, (1-\alpha)/6]$.

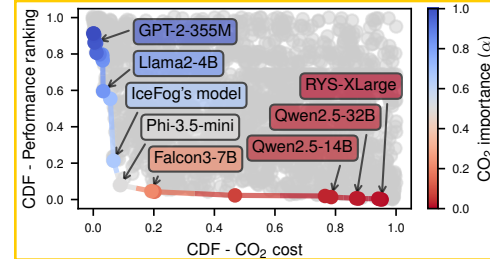


Figure 4: We can meaningfully explore the Pareto-optimal models of the Open LLM Leaderboard [12] with COPA. We use $p = \infty$ on the 7 objectives and highlight some of the selected models as we change the value of α .

We highlight the selected LLMs in Fig. 4, which groups all benchmarks into one dimension as their ∞ -norm for visualization purposes. We observe that the proposed norm enables the meaningful exploration of the Pareto front, with the values of α being uniformly spread-out across the front. Furthermore, not only can we sensibly explore the LLM space, but COPA enables interpreting these models in terms of the original objectives *and* the population they live in. For example, we can say that GPT-2 is Pareto-optimal as it consumes the least, but it only achieves a 6% average performance score, or that Phi-3.5-mini is a top 10% model in both aspects, consuming 0.53 kg of CO_2 vs. the 13 kg consumed by the best-performing model.

2. The fairness-accuracy trade-off. Moving to a more classic example, we consider how a DM could use COPA to choose a trade-off between accuracy and fairness in a classification problem, two objectives which are defined in completely different ways [62].

We reproduce the CelebA [34] experiment from Maheshwari and Perrot [37] using FairGrad—an algorithm whose hyperparameter ϵ upper-bounds the unfairness of the classifier—and create a population of models by sweeping through values of ϵ and five random initializations.

Fig. 5 (left) shows the Pareto front in the accuracy-fairness space, as we navigate it by changing α , clearly showing the difference between both objectives. Note that directly solving Eq. 3 leads to the solution with maximum accuracy, as in §5.1. Instead, using COPA we can uniformly navigate the Pareto front where, e.g., the robust min-max solution ($\alpha = 1/2$) lies precisely in the middle of the front. As a result, COPA offers a more reliable interpretation of its parameters than the upper-bound given by ϵ , which is clear by observing that, e.g., a value of $\epsilon = 1$ or 0.25 yields relatively similar solutions in Fig. 5.

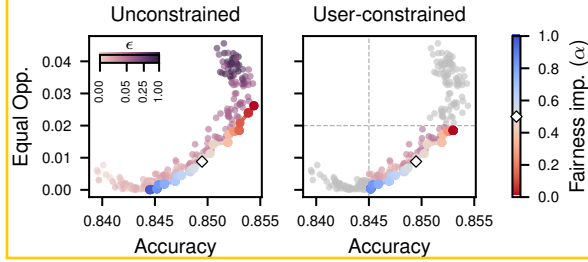


Figure 5: COPA can be used to meaningfully explore the Pareto front between accuracy and fairness (equal opportunity) in the CelebA experiment from Maheshwari and Perrot [37] in unconstrained (left) as well as user-constrained scenarios (right).

In addition, we consider a more realistic scenario where DMs bargain on acceptable values for the objectives, e.g., a regulatory body could demand equal opportunity to never exceed 0.02 [36]. Despite constraining the Pareto front to consider only valid solutions (we still use invalid ones to approximate the CDF), COPA stills provides a sensible way to navigate the space of valid models, proving that we can easily combine rules on the original and CDF-transformed objective spaces.

5.3 Case 2: Comparative model analysis

Previously, we have explored how DMs can meaningfully explore the Pareto front. Now, we focus on a related but different question: *How much could semantic incomparability alter the conclusions we draw from comparative analyses in ML research?*

1. Incomparable objectives. First, we consider a multitask learning (MTL) setting, where the heterogeneity of the tasks to solve makes it prone to face incomparable objectives. In fact, it is common to aggregate objectives with the average relative performance, Δ , as discussed in §4. To clearly showcase the issue, we look at the multi-SVHN experiment from Javaloy and Valera [25], which uses a modified version of SVHN [45] with a digit on each side of the image, and where we solve three classification tasks: **i)** left digit; **ii)** right digit; and **iii)** parity of their product; and two regression tasks: **iv)** sum of digits; and **v)** number of active pixels in the image.

Fig. 6 shows the ranking of the 14 MTL methods considered by Javaloy and Valera [25], if we were to use different criterion functions, namely: COPA with different values of p and equal weights, the average relative performance, Δ , and the regression error over the density task. The first two columns of the plot make extremely clear how much the density task dominates the average relative performance, perfectly matching its ranking. Again, this is a result of the reference method having nearly zero regression error on this task, greatly magnifying its relative performance, Δ_k .

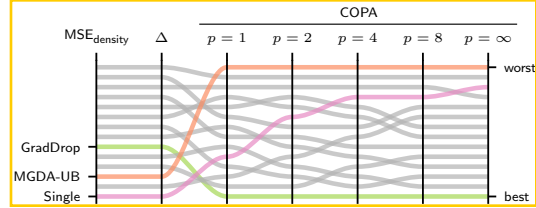


Figure 6: Ranking of MTL methods using different criteria to evaluate them. Methods whose rankings drastically change with Δ are colored.

As expected, the outlined issue has a tremendous impact on the conclusions drawn, e.g.: **i)** the worst method for all COPA instances, MGDA-UB [50], becomes the 3rd best method w.r.t. Δ ; or **ii)** the best one for every COPA, GradDrop [8], becomes the 6th best. Fig. 6 also shows that the reference method (Single) is among the least robust models ($p = \infty$), and slowly improves as we look less at individual performances ($p = 1$). It is worth-noting that the authors were aware of the issue and left the density task out when aggregating objectives, reporting both Δ and density MSE as a pair.

2. Seemingly comparable objectives. Sometimes, semantic incomparability can arise in unexpected scenarios. We take domain generalization as an example and, in particular, the DomainBed [17] experiment from Hemati et al. [19]. Here, the authors compare different methods by training them on some domains, and testing them on 4 unseen ones, reporting the average domain accuracy as commonly done in the literature.

Fig. 7 shows the ranking of the considered methods as we use different criterion functions, with the average accuracy in the first column. For two of the highlighted methods, RSC [22] and SagNet

[41], we observe their performance deteriorate and improve, respectively, as we consider less robust criteria, in accordance with the average accuracy. However, we see a different story with HGP [19] and Mixup [59], whose rankings are consistent for all COPA instances, but drastically change when we average accuracies. This leads to significantly different analyses concluding, e.g., that Mixup is worse than SagNet and HGP, in disagreement with every other criterion function.

In fact, accuracies present *significantly different ranges* across domains (see Tab 2) and differences in domains with less variance are less important in the average computation. If we normalize the results using norm_k (Eq. 6), we see that Mixup significantly outperforms HGP in these domains, swapping their rankings. This can also be observed in Fig. 7, where norm aligns much better with COPA.

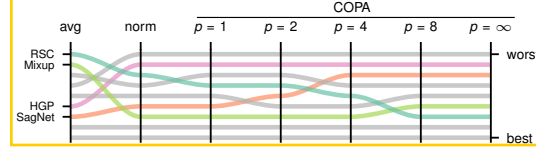


Figure 7: Ranking of domain generalization methods as we change the criterion function. Average accuracy is inconsistent with every COPA instance.

5.4 Case 3: Benchmarking

Finally, we motivate the use of COPA and CDF-normalized objectives in general benchmarking where, in contrast with the previous use cases, objectives are not necessarily aggregated into a scalar value, but plotted together. Additional plots can be found in §A.5.

We take the AutoML Benchmark (AMLB) [15] for the use-case as it “follows best practices and avoids common mistakes when comparing frameworks.” We reproduce all figures from the original work, comparing 15 AutoML methods evaluated on 104 different objectives. Since objectives are incomparable, the authors scale them using norm_k (Eq. 6) with a random forest as reference model, providing a number of analyses from these objectives. Remarkably, the authors also encourage the use of CD diagrams and Friedman tests, two methods that based on relative rankings.

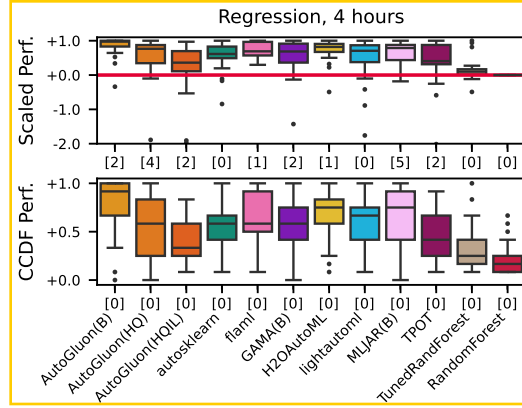


Figure 8: Comparison of AutoML methods on AMLB [15] using scaled performance, norm_k , with a random forest as reference method (red line); and using a CCDF-transformation (bottom). Brackets indicate the number of off-view outliers.

A natural step is therefore to use CDF-normalized objectives. Fig. 8 shows the same AMLB box-plot using scaled and CCDF performance, i.e., $1 - F_k(y_k)$. We find that using CCDFs comes with several benefits: *i)* there are no outliers to report, unlike in the original plot (all values lie in $[0, 1]$); *ii)* there is no need for an arbitrary reference model; and *iii)* we can provide clear population-based interpretations, e.g., “on average, AutoGluon(B) [10] yields over top-10% performance on the considered objectives.” These benefits extend to all AMLB plots, demonstrating that the proposed CDF transformation is a sensible way of normalizing objectives in general.

6 Concluding remarks

In this work, we have shown the importance of meaningfully navigating the Pareto front in multi-objective ML evaluation, allowing users to perform better-informed decisions. To this end, we have highlighted how crucial is to properly normalize all objectives and to have a criterion function that sensibly reflects DM preferences into an optimization problem. Finally, we have implemented these insights in COPA, and extensively demonstrated the impact that it can have in areas as fundamental and timely as model selection and benchmarking.

Our work opens many intriguing venues for future research. For example, we would be excited to see COPA adapted to active settings with humans-in-the-loop, criterion functions that parametrize other preference types, a formal systematization of model selection enabled by COPA, or its adoption in public portals such as the Open LLM Leaderboard [12] or the DecodingTrust benchmark [57].

Bibliography

- [1] Tadas Baltrušaitis, Chaitanya Ahuja, and Louis-Philippe Morency. Multimodal machine learning: A survey and taxonomy. *IEEE transactions on pattern analysis and machine intelligence*, 41(2):423–443, 2018. (Cited in page 6.)
- [2] Grigory Isaakovich Barenblatt. *Dimensional analysis*. CRC Press, 1987. (Cited in pages 3 and 6.)
- [3] V. Joseph Bowman. On the Relationship of the Tchebycheff Norm and the Efficient Frontier of Multiple-Criteria Objectives. In Hervé Thiriez and Stanley Zionts, editors, *Multiple Criteria Decision Making*, pages 76–86, Berlin Heidelberg, 1976. Springer Berlin Heidelberg. ISBN 978-3-642-87563-2. (Cited in page 3.)
- [4] Iuergen Branke, Kalyan Deb, Kaisa Miettinen, and Slowinski Roman. Multiobjective Optimization, Interactive and Evolutionary Approaches [outcome of Dagstuhl seminars]. 2008. (Cited in page 3.)
- [5] Rich Caruana and Alexandru Niculescu-Mizil. Data mining in metric space: an empirical analysis of supervised learning performance criteria. In *Proceedings of the tenth ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 69–78, 2004. (Cited in page 2.)
- [6] George Casella and Roger L Berger. *Statistical inference*. Cengage Learning, 2021. (Cited in pages 4 and 5.)
- [7] Weiyu Chen and James T. Kwok. Pareto Merging: Multi-Objective Optimization for Preference-Aware Model Merging. 2024. URL <https://api.semanticscholar.org/CorpusID:271924011>. (Cited in page 6.)
- [8] Zhao Chen, Jiquan Ngiam, Yanping Huang, Thang Luong, Henrik Kretzschmar, Yuning Chai, and Dragomir Anguelov. Just Pick a Sign: Optimizing Deep Multitask Models with Gradient Sign Dropout. In Hugo Larochelle, Marc’Aurelio Ranzato, Raia Hadsell, Maria-Florina Balcan, and Hsuan-Tien Lin, editors, *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*, 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/16002f7a455a94aa4e91cc34ebdb9f2d-Abstract.html>. (Cited in page 8.)
- [9] Tristan Coignion, Clément Quinton, and Romain Rouvoy. Green My LLM: Studying the key factors affecting the energy consumption of code assistants. *ArXiv preprint*, abs/2411.11892, 2024. URL <https://arxiv.org/abs/2411.11892>. (Cited in pages 1 and 7.)
- [10] Nick Erickson, Jonas Mueller, Alexander Shirkov, Hang Zhang, Pedro Larroy, Mu Li, and Alexander Smola. Autogluon-tabular: Robust and accurate automl for structured data. *ArXiv preprint*, abs/2003.06505, 2020. URL <https://arxiv.org/abs/2003.06505>. (Cited in page 9.)
- [11] Peter A. Flach. *ROC Analysis*, pages 869–875. Springer US, Boston, MA, 2010. ISBN 978-0-387-30164-8. doi: 10.1007/978-0-387-30164-8_733. URL https://doi.org/10.1007/978-0-387-30164-8_733. (Cited in page 6.)
- [12] Clémentine Fourier, Nathan Habib, Alina Lozovskaya, Konrad Szafer, and Thomas Wolf. Open LLM Leaderboard v2, 2024. URL https://huggingface.co/spaces/open-llm-leaderboard/open_llm_leaderboard. (Cited in pages 1, 2, 7, 9, 16, and 17.)
- [13] Milton Friedman. The Use of Ranks to Avoid the Assumption of Normality Implicit in the Analysis of Variance. *Journal of the American Statistical Association*, 32(200):675–701, 1937. ISSN 01621459, 1537274X. URL <http://www.jstor.org/stable/2279372>. (Cited in page 6.)
- [14] Gerry Geenens. (Re-)Reading Sklar (1959)—A Personal View on Sklar’s Theorem. *Mathematics*, 12(3):380, 2024. (Cited in pages 4 and 6.)
- [15] Pieter Gijsbers, Marcos L. P. Bueno, Stefan Coors, Erin LeDell, Sébastien Poirier, Janek Thomas, Bernd Bischl, and Joaquin Vanschoren. AMLB: an AutoML Benchmark. *Journal of Machine Learning Research*, 25(101):1–65, 2024. URL <http://jmlr.org/papers/v25/22-0493.html>. (Cited in pages 2, 9, 20, 21, and 22.)

- [16] Ian J. Goodfellow, Yoshua Bengio, and Aaron C. Courville. *Deep Learning*. Adaptive computation and machine learning. MIT Press, 2016. ISBN 978-0-262-03561-3. URL <http://www.deeplearningbook.org/>. (Cited in page 6.)
- [17] Ishaan Gulrajani and David Lopez-Paz. In Search of Lost Domain Generalization. In *9th International Conference on Learning Representations, ICLR 2021, Virtual Event, Austria, May 3-7, 2021*. OpenReview.net, 2021. URL <https://openreview.net/forum?id=1QdXeXDoWtI>. (Cited in page 8.)
- [18] David J Hand. Measuring classifier performance: a coherent alternative to the area under the ROC curve. *Machine learning*, 77(1):103–123, 2009. (Cited in page 6.)
- [19] Sobhan Hemati, Guojun Zhang, Amir Hossein Estiri, and Xi Chen. Understanding Hessian Alignment for Domain Generalization. In *IEEE/CVF International Conference on Computer Vision, ICCV 2023, Paris, France, October 1-6, 2023*, pages 18958–18968. IEEE, 2023. doi: 10.1109/ICCV51070.2023.01742. URL <https://doi.org/10.1109/ICCV51070.2023.01742>. (Cited in pages 8, 9, 18, 19, and 20.)
- [20] Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring Mathematical Problem Solving With the MATH Dataset, 2021. URL <https://arxiv.org/abs/2103.03874>. (Cited in pages 7 and 17.)
- [21] Dong Huang, Qingwen Bu, Jie Zhang, Xiaofei Xie, Junjie Chen, and Heming Cui. Bias assessment and mitigation in llm-based code generation. *ArXiv preprint*, abs/2309.14345, 2023. URL <https://arxiv.org/abs/2309.14345>. (Cited in page 1.)
- [22] Zeyi Huang, Haohan Wang, Eric P Xing, and Dong Huang. Self-challenging improves cross-domain generalization. In *Computer vision—ECCV 2020: 16th European conference, Glasgow, UK, August 23–28, 2020, proceedings, part II 16*, pages 124–140. Springer, 2020. (Cited in page 8.)
- [23] Amin Ibrahim, Azam Asilian Bidgoli, Shahryar Rahnamayan, and Kalyanmoy Deb. A Novel Pareto-optimal Ranking Method for Comparing Multi-objective Optimization Algorithms. *ArXiv preprint*, abs/2411.17999, 2024. URL <https://arxiv.org/abs/2411.17999>. (Cited in page 6.)
- [24] Nathalie Japkowicz and Mohak Shah. *Evaluating learning algorithms: a classification perspective*. Cambridge University Press, 2011. (Cited in pages 1 and 6.)
- [25] Adrián Javaloy and Isabel Valera. RotoGrad: Gradient Homogenization in Multitask Learning. In *The Tenth International Conference on Learning Representations, ICLR 2022, Virtual Event, April 25-29, 2022*. OpenReview.net, 2022. URL <https://openreview.net/forum?id=T8wHz4rnuGL>. (Cited in pages 8 and 18.)
- [26] Adrián Javaloy, Maryam Meghdadi, and Isabel Valera. Mitigating Modality Collapse in Multimodal VAEs via Impartial Optimization. In Kamalika Chaudhuri, Stefanie Jegelka, Le Song, Csaba Szepesvári, Gang Niu, and Sivan Sabato, editors, *International Conference on Machine Learning, ICML 2022, 17-23 July 2022, Baltimore, Maryland, USA*, volume 162 of *Proceedings of Machine Learning Research*, pages 9938–9964. PMLR, 2022. URL <https://proceedings.mlr.press/v162/javaloy22a.html>. (Cited in page 6.)
- [27] Peter Kairouz, H. Brendan McMahan, Brendan Aven, Aurélien Bellet, Mehdi Bennis, Arjun Nitin Bhagoji, Kallista Bonawitz, Zachary Charles, Graham Cormode, Rachel Cummings, Rafael G. L. D’Oliveira, Hubert Eichner, Salim El Rouayheb, David Evans, Josh Gardner, Zachary Garrett, Adrià Gascón, Badi Ghazi, Phillip B. Gibbons, Marco Gruteser, Zaid Harchaoui, Chaoyang He, Lie He, Zhouyuan Huo, Ben Hutchinson, Justin Hsu, Martin Jaggi, Tara Javidi, Gauri Joshi, Mikhail Khodak, Jakub Konečný, Aleksandra Korolova, Farinaz Koushanfar, Sanmi Koyejo, Tancrede Lepoint, Yang Liu, Prateek Mittal, Mehryar Mohri, Richard Nock, Ayfer Özgür, Rasmus Pagh, Hang Qi, Daniel Ramage, Ramesh Raskar, Mariana Raykova, Dawn Song, Weikang Song, Sebastian U. Stich, Ziteng Sun, Ananda Theertha Suresh, Florian Tramèr, Praneeth Vepakomma, Liyuan Wang, Li Xiong, Zheng Xu, Qiang Yang, Felix X. Yu, Han Yu, and Sen Zhao. Advances and Open Problems in Federated Learning. *Foundations and Trends® in Machine Learning*, 14(1–2):1–210, 2021. ISSN 1935-8237. doi: 10.1561/22000000083. URL <http://dx.doi.org/10.1561/22000000083>. (Cited in page 6.)
- [28] M. G. Kendall. A new measure of rank correlation. *Biometrika*, 30(1-2):81–93, 1938. ISSN 0006-3444. doi: 10.1093/biomet/30.1-2.81. URL <https://doi.org/10.1093/biomet/30.1-2.81>. (Cited in page 6.)

- [29] M. G. Kendall and R. M. Sundrum. Distribution-Free Methods and Order Properties. *Revue de l'Institut International de Statistique / Review of the International Statistical Institute*, 21(3): 124–134, 1953. ISSN 03731138. URL <http://www.jstor.org/stable/1401424>. (Cited in page 4.)
- [30] David Krueger, Ethan Caballero, Jörn-Henrik Jacobsen, Amy Zhang, Jonathan Binas, Dinghuai Zhang, Rémi Le Priol, and Aaron C. Courville. Out-of-Distribution Generalization via Risk Extrapolation (REx). In Marina Meila and Tong Zhang, editors, *Proceedings of the 38th International Conference on Machine Learning, ICML 2021, 18-24 July 2021, Virtual Event*, volume 139 of *Proceedings of Machine Learning Research*, pages 5815–5826. PMLR, 2021. URL <http://proceedings.mlr.press/v139/krueger21a.html>. (Cited in page 19.)
- [31] Saku Kukkonen and Jouni Lampinen. Ranking-dominance and many-objective optimization. In *2007 IEEE Congress on Evolutionary Computation*, pages 3983–3990. IEEE, 2007. (Cited in page 6.)
- [32] Lu Li, Tianyu Zhang, Zhiqi Bu, Suyuchen Wang, Huan He, Jie Fu, Yonghui Wu, Jiang Bian, Yong Chen, and Yoshua Bengio. MAP: Low-compute model merging with amortized Pareto fronts via quadratic approximation. *ArXiv preprint*, abs/2406.07529, 2024. URL <https://arxiv.org/abs/2406.07529>. (Cited in page 6.)
- [33] Bo Liu, Yihao Feng, Peter Stone, and Qiang Liu. FAMO: Fast Adaptive Multitask Optimization. In Alice Oh, Tristan Naumann, Amir Globerson, Kate Saenko, Moritz Hardt, and Sergey Levine, editors, *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10-16, 2023*, 2023. URL http://papers.nips.cc/paper_files/paper/2023/hash/b2fe1ee8d936ac08dd26f2ff58986c8f-Abstract-Conference.html. (Cited in pages 4 and 6.)
- [34] Ziwei Liu, Ping Luo, Xiaogang Wang, and Xiaoou Tang. Deep Learning Face Attributes in the Wild. In *2015 IEEE International Conference on Computer Vision, ICCV 2015, Santiago, Chile, December 7-13, 2015*, pages 3730–3738. IEEE Computer Society, 2015. doi: 10.1109/ICCV.2015.425. URL <https://doi.org/10.1109/ICCV.2015.425>. (Cited in pages 7 and 17.)
- [35] Alexandra Sasha Luccioni, Sylvain Viguier, and Anne-Laure Ligozat. Estimating the carbon footprint of bloom, a 176b parameter language model. *Journal of Machine Learning Research*, 24(253):1–15, 2023. (Cited in page 1.)
- [36] Mark MacCarthy. Standards of fairness for disparate impact assessment of big data algorithms. *Cumb. L. Rev.*, 48:67, 2017. (Cited in page 8.)
- [37] Gaurav Maheshwari and Michaël Perrot. FairGrad: Fairness Aware Gradient Descent. *ArXiv preprint*, abs/2206.10923, 2022. URL <https://arxiv.org/abs/2206.10923>. (Cited in pages 7, 8, and 17.)
- [38] Kevis-Kokitsi Maninis, Ilija Radosavovic, and Iasonas Kokkinos. Attentive Single-Tasking of Multiple Tasks. In *IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2019, Long Beach, CA, USA, June 16-20, 2019*, pages 1851–1860. Computer Vision Foundation / IEEE, 2019. doi: 10.1109/CVPR.2019.00195. URL http://openaccess.thecvf.com/content_CVPR_2019/html/Maninis_Attentive_Single-Tasking_of_Multiple_Tasks_CVPR_2019_paper.html. (Cited in page 4.)
- [39] Natalia Martínez, Martín Bertrán, and Guillermo Sapiro. Minimax Pareto Fairness: A Multi Objective Perspective. In *Proceedings of the 37th International Conference on Machine Learning, ICML 2020, 13-18 July 2020, Virtual Event*, volume 119 of *Proceedings of Machine Learning Research*, pages 6755–6764. PMLR, 2020. URL <http://proceedings.mlr.press/v119/martinez20a.html>. (Cited in page 6.)
- [40] Kaisa Miettinen. *Nonlinear multiobjective optimization*, volume 12. Springer Science & Business Media, 1999. (Cited in pages 4 and 6.)
- [41] Jun Hyun Nam, Hyuntak Cha, Sungsoo Ahn, Jaeho Lee, and Jinwoo Shin. Learning from Failure: De-biasing Classifier from Biased Classifier. In Hugo Larochelle, Marc’Aurelio Ranzato, Raia Hadsell, Maria-Florina Balcan, and Hsuan-Tien Lin, editors, *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*, 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/eddc3427c5d77843c2253f1e799fe933-Abstract.html>. (Cited in page 9.)

- [42] Humza Naveed, Asad Ullah Khan, Shi Qiu, Muhammad Saqib, Saeed Anwar, Muhammad Usman, Naveed Akhtar, Nick Barnes, and Ajmal Mian. A comprehensive overview of large language models. *ArXiv preprint*, abs/2307.06435, 2023. URL <https://arxiv.org/abs/2307.06435>. (Cited in page 7.)
- [43] Aviv Navon, Aviv Shamsian, Idan Achituve, Haggai Maron, Kenji Kawaguchi, Gal Chechik, and Ethan Fetaya. Multi-Task Learning as a Bargaining Game. In Kamalika Chaudhuri, Stefanie Jegelka, Le Song, Csaba Szepesvári, Gang Niu, and Sivan Sabato, editors, *International Conference on Machine Learning, ICML 2022, 17-23 July 2022, Baltimore, Maryland, USA*, volume 162 of *Proceedings of Machine Learning Research*, pages 16428–16446. PMLR, 2022. URL <https://proceedings.mlr.press/v162/navon22a.html>. (Cited in pages 1 and 6.)
- [44] Alfredo Nazabal, Pablo M Olmos, Zoubin Ghahramani, and Isabel Valera. Handling incomplete heterogeneous data using VAEs. *Pattern Recognition*, 107:107501, 2020. (Cited in pages 2 and 6.)
- [45] Yuval Netzer, Tao Wang, Adam Coates, Alessandro Bissacco, Bo Wu, and Andrew Y Ng. Reading digits in natural images with unsupervised feature learning. *NeurIPS Workshop on Deep Learning and Unsupervised Feature Learning*, 2011. (Cited in page 8.)
- [46] Ji Won Park, Natasa Tagasovska, Michael Maser, Stephen Ra, and Kyunghyun Cho. BOTied: Multi-objective Bayesian optimization with tied multivariate ranks. In *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*. OpenReview.net, 2024. URL <https://openreview.net/forum?id=cj5HbaX14p>. (Cited in page 6.)
- [47] Ashley Piggins. *Collective Choice and Social Welfare—Expanded Edition*, 2019. (Cited in page 6.)
- [48] Alexandre Ramé, Corentin Dancette, and Matthieu Cord. Fishr: Invariant Gradient Variances for Out-of-Distribution Generalization. In Kamalika Chaudhuri, Stefanie Jegelka, Le Song, Csaba Szepesvári, Gang Niu, and Sivan Sabato, editors, *International Conference on Machine Learning, ICML 2022, 17-23 July 2022, Baltimore, Maryland, USA*, volume 162 of *Proceedings of Machine Learning Research*, pages 18347–18377. PMLR, 2022. URL <https://proceedings.mlr.press/v162/rame22a.html>. (Cited in pages 1 and 6.)
- [49] David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. GPOA: A Graduate-Level Google-Proof O&A Benchmark, 2023. URL <https://arxiv.org/abs/2311.12022>. (Cited in pages 7 and 17.)
- [50] Ozan Sener and Vladlen Koltun. Multi-Task Learning as Multi-Objective Optimization. In Samy Bengio, Hanna M. Wallach, Hugo Larochelle, Kristen Grauman, Nicolò Cesa-Bianchi, and Roman Garnett, editors, *Advances in Neural Information Processing Systems 31: Annual Conference on Neural Information Processing Systems 2018, NeurIPS 2018, December 3-8, 2018, Montréal, Canada*, pages 525–536, 2018. URL <https://proceedings.neurips.cc/paper/2018/hash/432aca3a1e345e339f35a30c8f65edce-Abstract.html>. (Cited in page 8.)
- [51] M Sklar. Fonctions de répartition à n dimensions et leurs marges. In *Annales de l’ISUP*, volume 8, pages 229–231, 1959. (Cited in page 4.)
- [52] Zayne Sprague, Xi Ye, Kaj Bostrom, Swarat Chaudhuri, and Greg Durrett. MuSR: Testing the Limits of Chain-of-thought with Multistep Soft Reasoning. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024. URL <https://openreview.net/forum?id=jenyYQzue1>. (Cited in pages 7 and 17.)
- [53] Baochen Sun and Kate Saenko. Deep coral: Correlation alignment for deep domain adaptation. In *Computer Vision – ECCV 2016 Workshops: Amsterdam, The Netherlands, October 8-10 and 15-16, 2016, Proceedings, Part III 14*, pages 443–450. Springer, 2016. (Cited in page 19.)
- [54] Mirac Suzgun, Nathan Scales, Nathanael Schärli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc Le, Ed Chi, Denny Zhou, and Jason Wei. Challenging BIG-Bench Tasks and Whether Chain-of-Thought Can Solve Them. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, editors, *Findings of the Association for Computational Linguistics: ACL 2023*, pages 13003–13051. Toronto, Canada, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-acl.824. URL <https://aclanthology.org/2023.findings-acl.824>. (Cited in pages 7 and 17.)

- [55] Howard G Tucker. A generalization of the Glivenko-Cantelli theorem. *The Annals of Mathematical Statistics*, 30(3):828–830, 1959. (Cited in page 5.)
- [56] Sergio Verdu and H Poor. On minimax robustness: A general approach and applications. *IEEE transactions on Information Theory*, 30(2):328–340, 1984. (Cited in pages 4 and 6.)
- [57] Boxin Wang, Weixin Chen, Hengzhi Pei, Chulin Xie, Mintong Kang, Chenhui Zhang, Chejian Xu, Zidi Xiong, Ritik Dutta, Rylan Schaeffer, Sang T. Truong, Simran Arora, Mantas Mazeika, Dan Hendrycks, Zinan Lin, Yu Cheng, Sanmi Koyejo, Dawn Song, and Bo Li. DecodingTrust: A Comprehensive Assessment of Trustworthiness in GPT Models. In Alice Oh, Tristan Naumann, Amir Globerson, Kate Saenko, Moritz Hardt, and Sergey Levine, editors, *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*, 2023. URL http://papers.nips.cc/paper_files/paper/2023/hash/63cb9921eecf51bfad27a99b2c53dd6d-Abstract-Datasets_and_Benchmarks.html. (Cited in pages 2, 6, and 9.)
- [58] Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhui Chen. MMLU-Pro: A More Robust and Challenging Multi-Task Language Understanding Benchmark. In Amir Globersons, Lester Mackey, Danielle Belgrave, Angela Fan, Ulrich Paquet, Jakub M. Tomczak, and Cheng Zhang, editors, *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*, 2024. URL http://papers.nips.cc/paper_files/paper/2024/hash/ad236edc564f3e3156e1b2feafb99a24-Abstract-Datasets_and_Benchmarks_Track.html. (Cited in pages 7 and 17.)
- [59] Yufei Wang, Haoliang Li, and Alex C. Kot. Heterogeneous Domain Generalization Via Domain Mixup. In *2020 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2020, Barcelona, Spain, May 4-8, 2020*, pages 3622–3626. IEEE, 2020. doi: 10.1109/ICASSP40776.2020.9053273. URL <https://doi.org/10.1109/ICASSP40776.2020.9053273>. (Cited in page 9.)
- [60] Frank Wilcoxon. Individual Comparisons by Ranking Methods. *Biometrics Bulletin*, 1(6):80–83, 1945. ISSN 00994987. URL <http://www.jstor.org/stable/3001968>. (Cited in page 6.)
- [61] Lifan Yuan, Yangyi Chen, Ganqu Cui, Hongcheng Gao, Fangyuan Zou, Xingyi Cheng, Heng Ji, Zhiyuan Liu, and Maosong Sun. Revisiting Out-of-distribution Robustness in NLP: Benchmarks, Analysis, and LLMs Evaluations. In Alice Oh, Tristan Naumann, Amir Globerson, Kate Saenko, Moritz Hardt, and Sergey Levine, editors, *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*, 2023. URL http://papers.nips.cc/paper_files/paper/2023/hash/b6b5f50a2001ad1cbccca96e693c4ab4-Abstract-Datasets_and_Benchmarks.html. (Cited in page 1.)
- [62] Muhammad Bilal Zafar, Isabel Valera, Manuel Gomez-Rodriguez, and Krishna P. Gummadi. Fairness Constraints: Mechanisms for Fair Classification. In Aarti Singh and Xiaojin (Jerry) Zhu, editors, *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics, AISTATS 2017, 20-22 April 2017, Fort Lauderdale, FL, USA*, volume 54 of *Proceedings of Machine Learning Research*, pages 962–970. PMLR, 2017. URL <http://proceedings.mlr.press/v54/zafar17a.html>. (Cited in pages 2 and 7.)
- [63] Milan Zeleny. Compromise programming. *Multiple criteria decision making*, 1973. (Cited in page 3.)
- [64] Yifan Zhong, Chengdong Ma, Xiaoyuan Zhang, Ziran Yang, Haojun Chen, Qingfu Zhang, Siyuan Qi, and Yaodong Yang. Panacea: Pareto Alignment via Preference Adaptation for LLMs. In Amir Globersons, Lester Mackey, Danielle Belgrave, Angela Fan, Ulrich Paquet, Jakub M. Tomczak, and Cheng Zhang, editors, *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*, 2024. URL http://papers.nips.cc/paper_files/paper/2024/hash/89f39d0b3d49a47606a165eefba2778c-Abstract-Conference.html. (Cited in page 6.)

709	[65]	Jeffrey Zhou, Tianjian Lu, Swaroop Mishra, Siddhartha Brahma, Sujoy Basu, Yi Luan, Denny
710		Zhou, and Le Hou. Instruction-Following Evaluation for Large Language Models, 2023. URL
711		https://arxiv.org/abs/2311.07911 . (Cited in pages 7 and 17.)

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